

The Relationship Between Board Certification and Disciplinary Actions Against Board-Eligible Family Physicians

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Abstract

Purpose

Lack of specialty board certification has been reported as a significant physician-level predictor of receiving a disciplinary action from a state medical board.

This study investigated the association between family physicians receiving a disciplinary action from a state medical board and certification by the American Board of Family Medicine (ABFM).

Method

Three datasets were merged and a series of logistic regressions were conducted examining the relationship between certification status and disciplinary actions

when adjusting for covariates. Data were available from 1976 to 2017. Predictor variables were gender, age, medical training degree type, medical school location, and the severity of the action.

Results

Of the family physicians in this sample, 95% (114,454/120,443) had never received any disciplinary action. Having ever been certified was associated with a reduced likelihood of ever receiving an action (odds ratio [OR] = 0.35; 95% confidence interval [CI] = 0.30, 0.40; $P < .001$), and having held a prior but not current certification at the time

of the action was associated with an increase in receiving the most severe type of action (OR = 3.71; 95% CI = 2.24, 6.13; $P < .001$).

Conclusions

Disciplinary actions are uncommon events. Family physicians who had ever been ABFM certified were less likely to receive an action. The most severe actions were associated with decreased odds of being board certified at the time of the action. Receiving the most severe action type increased the likelihood of physicians holding a prior but not current certification.

A medical license is a legal authorization from a state or territorial medical board that allows a physician to practice medicine in that jurisdiction and assures patients that they are receiving treatment from a trained professional.¹ State medical boards are also empowered to revoke the license of physicians when they violate professional standards of practice in that state.² Whereas a license to practice medicine is an undifferentiated license—physicians are not limited by their license in their scope of practice—medical specialty board certification is intended to signify a physician's specific training and demonstrated skills in a particular medical or surgical specialty or subspecialty.

To Err Is Human: Building a Safer Health System, the influential report from the Institute of Medicine (now the National Academy of Medicine), declared that a small proportion of the physician population “may be incompetent, impaired, uncaring, or may even have criminal intent.”³ Previous research has shown that a small number of physicians are responsible for a large proportion of negative patient outcomes. In the United States, 1% of all physicians accounted for 32% of all paid malpractice claims⁴; in Australia, 3% of the medical workforce accounted for 49% of patient complaints⁵; and in Canada, 10% of physicians accounted for 20% of licensure actions.⁶

Previous studies have reported that significant physician-level predictors of medical license disciplinary action in the United States include lack of specialty board certification,^{7–9} practicing family medicine,^{7,10} male gender,^{7,8,11} practicing more than 20 years,^{8,11} and international medical graduate (IMG) status.^{7,12} Although male gender was found to be a significant predictor in some studies,^{7,8,11} it was not found to be significant in another.¹⁰ One study found that women

were disciplined less often but more severely.⁸ Limitations of previous research have included studies of only a single state licensing board,^{7,8,11} a small selection of medical school graduates,^{10,11} or internal medicine residents only.⁹ Although research has shown that family physicians are more likely to receive disciplinary actions from state medical boards, to our knowledge no national study has explored the relationship between disciplinary action and board certification in family medicine.

To examine the association between American Board of Family Medicine (ABFM) certification and disciplinary action against a physician from a state medical board, we looked at whether ABFM-certified physicians received fewer disciplinary actions than non-ABFM-certified family physicians. We controlled for demographic variables such as gender and age to examine how other family physician characteristics influenced their likelihood of receiving a disciplinary action. We also examined the association between the severity of a disciplinary action with the likelihood that these physicians held a current or prior certification at the time of the action.

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Data

We excluded osteopathic physicians who completed AOA-accredited training who appeared in the AMA Physician Masterfile but did not appear in the ABFM dataset ($n = 2,379$). This exclusion was incorporated to better identify those osteopathic physicians who would have had the opportunity (based on completing a dual ACGME/AOA residency program) to elect to certify with the ABFM but did not do so. Prior to 2005, those osteopathic physicians

We first calculated bivariate statistics comparing differences in physician demographics according to whether or not a physician had ever received a disciplinary action from a state medical board, using *t* tests and chi-square analyses where applicable. Next, we examined the extent to which family physicians received disciplinary actions from state medical boards by computing

Lastly, we used logistic regression to examine the relationship between severity of the disciplinary action and prior certification for physicians who were not certified when they received a disciplinary action but did hold an initial certification at some point in the past. Predictor variables were gender, medical training degree type, medical school location, age by decade at the time of the action, and the severity of the action.

Results

Available physician demographic information is presented in Table 1. The majority of family physicians in the sample were male (61.2%, $n = 73,172$), held an MD degree (89.7%, $n = 107,181$), were graduates of U.S. medical schools (77.1%, $n = 92,839$), and never received a disciplinary action (95.0%, $n = 114,454$). The 5% ($n = 5,989$) of family physicians who did receive a disciplinary action were more likely to be male (84.0%, $n = 4,961$, $P < .001$), to hold an MD (92.7%, $n = 5,473$, $P < .001$), and to have received their action between the ages of 40 and 49 (33.3%, $n = 1,992$, $P < .001$).

Next, we calculated the number of actions against each physician. These actions could be from separate disciplinary events, multiple actions for the same event, or reciprocal actions by another state board for the same event. Of the 5.0% ($n = 5,989$) of physicians who had received a disciplinary action, 54.6% ($n = 3,269$) received only a single disciplinary action, 21.6% ($n = 1,291$) received two separate disciplinary actions, and 23.9% ($n = 1,429$) received three or more separate disciplinary actions (data not shown). The greatest number of actions incurred by a physician was 14.

The number of actions that occurred while a physician was certified is categorized by the severity of the action in Table 2. When a disciplinary action was taken, more than twice the number occurred while a physician was not certified ($n = 8,036$; 67.3%) than when a physician was certified ($n = 3,893$; 32.6%). More than six times as many cases of the most severe actions for physicians occurred for those who were not certified than for those who were certified ($n = 3,644$ vs. $n = 603$, respectively). In addition, twice as many of the somewhat severe actions were taken against noncertified physicians ($n = 3,667$; 45.6%) than the certified physicians ($n = 1,808$; 46.4%), while twice as many cases of the least severe actions were taken against the certified physicians ($n = 1,482$; 38.1%) than the noncertified physicians ($n = 725$; 9.0%).

Ever certified. Results of the logistic regression analyzing disciplinary action as a function of gender, degree type, medical school, and whether a physician was ever certified are shown in Table 3. Being male

Table 1

Demographics of Family Physicians Eligible for American Board of Family Medicine Certification in the United States, by Whether They Ever Received a Medical License Disciplinary Action, From a Study of Board Certification and Disciplinary Action, 2017

Characteristic	Received an action			Ever certified			Action while certified		
	No. (%) total ($N = 120,443$)	No. (%) no ($n = 114,454$)	No. (%) yes ($n = 5,989$)	P value	No.	No. (%) no ($n = 369$)	No. (%) yes ($n = 5,620$)	P value	No.
Gender^a									
Female	46,306 (38.8)	45,364 (39.9)	942 (16.0)		52 (18.4)	890 (15.8)			
Male	73,172 (61.2)	68,211 (60.1)	4,961 (84.0)		231 (81.6)	4,730 (84.2)			
				$< .001$	119,478			.292	5,903
Degree type^a									
DO	12,297 (10.3)	11,867 (10.4)	430 (7.28)		40 (14.1)	390 (6.94)			
MD	107,181 (89.7)	101,708 (89.6)	5,473 (92.7)		243 (85.9)	5,230 (93.1)			
				$< .001$	119,478			$< .001$	5,903
Medical school^b									
USMG	92,839 (77.1)	88,066 (77.0)	4,773 (79.8)		239 (66.2)	4,534 (80.7)			
IMG	27,552 (22.9)	26,345 (23.0)	1,207 (20.2)		122 (33.8)	1,085 (19.3)			
				$< .001$	120,391			$< .001$	5,980
Age decade									
< 40	895 (0.74)	—	895 (14.9)		79 (21.4)	802 (14.3)			
40–49	1,992 (1.65)	—	1,992 (33.3)		143 (38.8)	1,857 (33.0)			
50–59	1,830 (1.52)	—	1,830 (30.6)		99 (26.8)	1,745 (31.0)			
> 60 or greater	1,271 (1.06)	—	1,271 (21.2)		47 (12.7)	1,216 (21.6)			
				$< .001$	120,443			$< .001$	5,989
Certification									
Ever certified	117,046 (97.2)	111,426 (97.4)	5,620 (93.8)		—	—			
Never certified	3,397 (2.82)	3,028 (2.65)	369 (6.16)		—	—			
				$< .001$	120,443			$< .001$	5,989

Abbreviations: DO indicates doctor of osteopathy; MD, medical doctor; USMG, United States medical graduate; IMG, international medical graduate.

^a965 missing.

^b52 missing.

Table 2

Types of Disciplinary Actions Taken, by Certification Status, From a Study of Family Medicine Board Certification and Disciplinary Action, 2017

Severity	Action category ^a	No. (%) all (N = 11,929)	Action while certified	
			No. (%) no (n = 8,036)	No. (%) yes (n = 3,893)
Least		2,207 (18.5)	725 (9.02)	1,482 (38.1)
	CME required	384 (3.22)	95 (1.18)	289 (7.42)
	Fine	755 (6.33)	257 (3.20)	498 (12.8)
	Reprimand	1,068 (8.95)	373 (4.64)	695 (17.9)
Somewhat		5,475 (45.9)	3,667 (45.6)	1,808 (46.4)
	Probation	2,174 (18.2)	1,379 (17.2)	795 (20.4)
	Restricted	2,048 (17.2)	1,620 (20.2)	428 (11.0)
	Conditions	1,253 (10.5)	668 (8.31)	585 (15.0)
Most		4,247 (35.6)	3,644 (45.3)	603 (15.5)
	Revoked	768 (6.44)	650 (8.09)	118 (3.03)
	Surrendered	873 (7.32)	762 (9.48)	111 (2.85)
	Denied	318 (2.67)	249 (3.10)	69 (1.77)
	Suspension	2,288 (19.2)	1,983 (24.7)	305 (7.83)

Abbreviation: CME indicates continuing medical education.

^at tests for severity by action while certified and action category by action while certified were both statistically significant ($P < .001$, $N = 11,929$).

was associated with an increase in the odds of receiving a disciplinary action (odds ratio [OR] = 3.44; 95% confidence interval [CI] = 3.20, 3.69; $P < .001$), as was holding an MD (OR = 1.58; 95% CI = 1.43, 1.75; $P < .001$) and being a graduate of a U.S. medical school (OR = 1.22; 95% CI = 1.14, 1.31; $P < .001$). However, having ever been certified was associated with reduced odds of ever receiving a disciplinary action (OR = 0.35; 95% CI = 0.30, 0.40; $P < .001$).

Current certification. The results of the logistic regression exploring the

relationship between gender, degree type, medical school, age, and the severity of the disciplinary action and certification at time of disciplinary action are shown in Table 4. There was not a significant effect between gender and being certified at the time of a disciplinary action (OR = 0.88; 95% CI = 0.75, 1.03; $P = .011$). However, MDs (OR = 1.34; 95% CI = 1.07, 1.67; $P = .010$) and graduates of U.S. medical schools (OR = 1.27; 95% CI = 1.10, 1.48; $P = .001$) showed increased odds of being certified at the time of a disciplinary action. There was also not a significant

effect in the odds of physicians 40 to 49 years (OR = 0.98; 95% CI = 0.83, 1.17; $P = 0.851$) compared with those less than 40 years and being certified at time of action, though physicians who were 50 to 59 years old when they received their action had a decrease in the odds of being certified at the time of the action (OR = 0.80; 95% CI = 0.67, 0.95; $P = .014$), as did those who were 60 years old or greater when they received their action (OR = 0.48; 95% CI = 0.39, 0.59; $P < .001$). Finally, physicians with actions classified as somewhat or most severe, compared with least severe, had a continual decrease in the odds of being board certified at the time of an action (somewhat severe OR = 0.26; 95% CI = 0.23, 0.30; $P < .001$; most severe OR = 0.08; 95% CI = 0.07, 0.10; $P < .001$).

Prior but not current certification.

The results of the logistic regression exploring the relationship between gender, degree type, medical school, age, and severity of the disciplinary action and prior certification are shown in Table 5. Gender (OR = 1.19; 95% CI = 0.81, 1.74; $P = .369$) and degree type (OR = 1.00; 95% CI = 0.56, 1.71; $P = .986$) did not significantly change the odds of physicians having held prior certification in the past but not at the time of the disciplinary action; however, graduating from a U.S. medical school (OR = 2.79; 95% CI = 1.97, 3.94; $P < .001$) increased the odds of having held a prior certification at the time of the action. Increasing physician age was associated with a continual increase in the odds that they had prior certification at the time of the action (40–49 years OR = 10.44; 95% CI = 7.61, 14.45; $P < .001$; 50–59 years OR = 46.00; 95% CI = 28.76, 77.19; $P < .001$; 60 years old or greater OR = 434.71; 95% CI = 136.20, 2,651.70; $P < .001$). Finally, most severe compared with least severe actions were associated with an increase in the odds that physicians held a prior certification in the past but were not certified at the time the disciplinary action was incurred (OR = 3.71; 95% CI = 2.24, 6.13; $P < .001$).

Discussion

Findings from our study demonstrate that the overwhelming majority of family physicians, 95%, have never received a disciplinary action from their state medical board. When a disciplinary

Table 3

Association Between Ever Receiving a Disciplinary Action and Physician Characteristics (N = 120,443), From a Study of Family Medicine Board Certification and Disciplinary Action, 2017^a

Variable	OR	95% CI	P value
Gender (female)	Ref.		
Male	3.44	3.20, 3.69	< .001
Degree type (DO)	Ref.		
MD	1.58	1.43, 1.75	< .001
Medical school (IMG)	Ref.		
USMG	1.22	1.14, 1.31	< .001
Ever certified (no)	Ref.		
Yes	0.35	0.30, 0.40	< .001

Abbreviations: DO indicates doctor of osteopathy; MD, medical doctor; USMG, United States medical graduate; IMG, international medical graduate; OR, odds ratio; CI, confidence interval.

^aOnly adjusted results reported.

Table 4

Association Between Holding Certification at Time of License Action and Physician Characteristics and Severity of Action (N = 5,989), From a Study of Family Medicine Board Certification and Disciplinary Action, 2017^a

Variable	OR	95% CI	P value
Gender (female)	Ref.		
Male	0.88	0.75, 1.03	.011
Degree type (DO)	Ref.		
MD	1.34	1.07, 1.67	.010
Medical school (IMG)	Ref.		
USMG	1.27	1.10, 1.48	.001
Age decade (< 40)	Ref.		
40–49	0.98	0.83, 1.17	.851
50–59	0.80	0.67, 0.95	.014
> 60	0.48	0.39, 0.59	< .001
Severity (least)	Ref.		
Somewhat	0.26	0.23, 0.30	< .001
Most	0.08	0.07, 0.10	< .001

Abbreviations: DO indicates doctor of osteopathy; MD, medical doctor; USMG, United States medical graduate; IMG, international medical graduate; OR, odds ratio; CI, confidence interval.

^aOnly adjusted results reported.

action was taken, more than twice the number occurred while a physician was not certified than when a physician was certified, and more than six times as many disciplinary cases were classified as most severe for physicians who were not certified than for those who were certified.

Our study suggests that board certification is an important component

of public protection. The findings indicate that ever having been certified was associated with reduced odds of ever receiving a disciplinary action from a state or territorial medical board. In contrast, we found that those family physicians who had a most severe action compared with a least severe action had an increase in the odds that they held a prior ABFM certificate that had lapsed, suggesting a strong association

between severity of actions and continual certification status. It should be noted that a physician could have a lapsed certificate because of a previous license action, in which case the certificate would have been revoked, or for a personal decision to not continue with the certification program. Our purpose was not to differentiate between these two possibilities but, rather, to examine the effect of lapsed certificates for any reason. Our findings would seem to refute claims that initial board certification is sufficient for public protection.¹³

This study supports previous research suggesting that male physicians, generally, are more likely to receive disciplinary actions.^{7,8,11} Female physicians are reported to have a more patient-centered communication style,^{14–17} which may lead to fewer patient complaints. Female physicians have also been shown to have better patient outcomes in some settings,¹⁸ perhaps because they tend to adhere to clinical guidelines,^{19–21} and provide preventive care^{22–24} more often than their male counterparts.

Our findings do not support previous research suggesting that IMGs are disciplined more often than U.S. medical school graduates.^{7,12} We found the opposite—that graduating from a U.S. medical school was associated with increased odds of receiving a disciplinary action; however, we have no theories to account for this finding, and additional research should be conducted to investigate this relationship.

Previous research has focused simply on whether older physicians are more likely to receive disciplinary actions, but we focused on the physician's age at the time a disciplinary action occurred and found that physicians who were 50 years or older when they received such actions were less likely to be certified at the time of the action. A quick calculation of the unadjusted analysis from Table 1 shows that 96% of physicians over 60 were certified at some point (1,216/1,263), but only 30% were certified when they received their action (374/1,263). For comparison, 91% (802/881) of those younger than 40 had ever been certified, but 47% (417/881) were certified when they received their action. This suggests that engaging in a continuous certification program may be especially important as physicians age.^{8,11}

Table 5

Association Between Having Had a Prior but Not Current Certification at Time of License Action and Physician Characteristics and Severity of Action (N = 3,494), From a Study of Family Medicine Board Certification and Disciplinary Action, 2017^a

Variable	OR	95% CI	P value
Gender (female)	Ref.		
Male	1.19	0.81, 1.74	.369
Degree type (DO)	Ref.		
MD	1.00	0.56, 1.71	.986
Medical school (IMG)	Ref.		
USMG	2.79	1.97, 3.94	< .001
Age decade (< 40)	Ref.		
40–49	10.44	7.61, 14.45	< .001
50–59	46.00	28.76, 77.19	< .001
> 60	434.71	136.20, 2,651.70	< .001
Severity (least)	Ref.		
Somewhat	1.26	0.81, 1.94	.299
Most	3.71	2.24, 6.13	< .001

Abbreviations: DO, doctor of osteopathy; MD, medical doctor; USMG, United States medical graduate; IMG, international medical graduate; OR, odds ratio; CI, confidence interval.

^aOnly adjusted results reported.

Limitations

This study has several limitations. We eliminated osteopathic physicians listed in the AMA file who were not in the ABFM dataset to ensure that those who were included were eligible for ABFM certification. It is possible that through this approach we eliminated osteopathic physicians who were AOA certified or, alternately, those who did not have either American Board of Medical Specialties member board or AOA certification. Our eligibility criteria required that physicians' last residency program was completed in family medicine; this increased the likelihood that subjects were practicing primary care. However, it is possible that, for instance, someone first completing a three-year family medicine residency and then a three-year emergency medicine residency could be practicing as a family physician, but we assumed for the sake of this study that such a physician would likely be practicing emergency medicine. It is also possible that our eligibility criteria and subsequent match rate of 82.5% may have introduced some selection bias into the sample. Furthermore, our decision to select the most severe action may have skewed the results toward an inflated rate of severe actions. Finally, the AMA Masterfile contains self-reported data, and we did not independently verify data contained in it.

Conclusion

Receiving a disciplinary action against a medical license from a state medical board is an uncommon event. Family physicians who had ever been certified by the ABFM were less likely to ever receive a disciplinary action. If a physician did receive a disciplinary action, the most severe actions were associated with decreased odds of being board certified at the time of the action. Actions categorized as most severe and being older at the time of the action increased the likelihood that the physicians held a prior but not current certification.

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